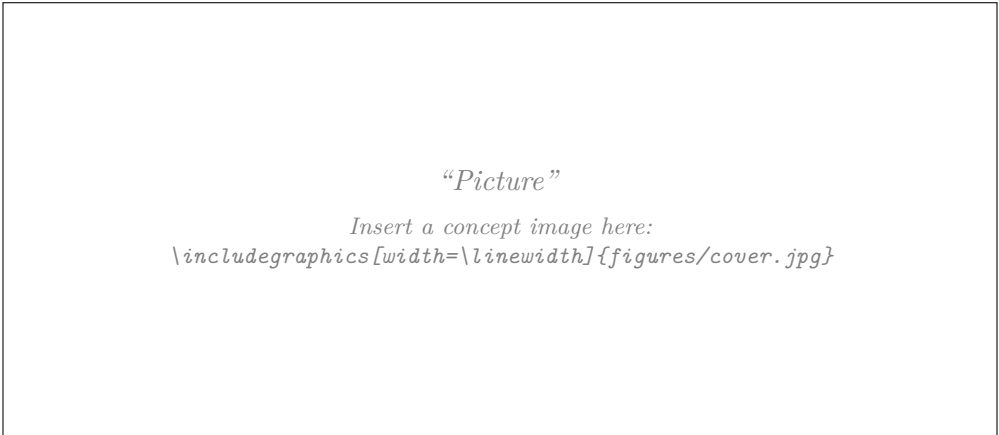


Report Biomimicry Challenge

ZHAW Institute for Sustainable Development | Blockweek Bionics 2026

Challenge 1 — Urban Heat Islands

Cooling the City



Students	Contribution to the report / (sections)
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[TODO: Name 3]	[TODO: e.g. Sections 1, 5, figures and editing]
[TODO: Name 4]	[TODO: e.g. Section 3.3, measurements, appendix]
Comments	[TODO: optional remarks for the instructors]

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1 Introduction

1.1 Background and relevance

Impervious urban surfaces absorb most of the incoming short-wave radiation and store it, sealing removes the evaporative cooling that vegetation would provide, and the narrow geometry of a street canyon keeps the emitted long-wave radiation from escaping to the sky (Oke 1982). Zurich measures the result as an air-temperature difference of 1 K to 2 K by day and of up to 7 K at night against the surrounding area, with sun-exposed sealed surfaces above 60 °C in summer (Stadt Zürich 2021). The stored energy is released into the street canyon, and that is where the load is felt: a surface at 60 °C radiates at eye level onto anyone walking, waiting or sitting in the street, while the same person indoors is shielded from it and can ventilate at night. Public outdoor space is the part of the city that cannot be conditioned, and it is where the city's health concern applies — cerebrovascular, cardiovascular and respiratory illness during heat periods, with older people and small children most at risk (Stadt Zürich 2021).

Air conditioning is no answer to this: it rejects the heat into the street as waste heat and raises electricity demand at the peak hour, which is why the city advises against it (Stadt Zürich 2020a, pp. 132–133). What Zurich pursues instead — the *Fachplanung Hitzeminderung* with its thematic maps and implementation agenda (Stadt Zürich 2020a; Stadt Zürich 2020b; Stadt Zürich 2020c) — acts on the outdoor space at pedestrian level. That space is bounded by pavement and by the façades of a building stock that already stands, so those surfaces are the intervention point a retrofit can reach: the target is the street in front of the façade, not the room behind it. Section 2.3 sets out what the city specifies for them today, and with it the benchmark this project has to beat.

1.2 Aim of the report and intended contribution

This report documents a biomimicry design process for reducing heat load on the envelope of existing urban buildings, and with it on the street space those envelopes enclose. It follows the Biomimicry Design Spiral (Baumeister et al. 2014) through scoping, biological discovery and abstraction, engineering concept development, validation and evaluation, in the challenge-to-biology direction described by Helms et al. (2009). The intended contribution is a technically specified, quantitatively checked envelope concept, together with a transparent account of which biological mechanisms were transferred, which were rejected, and on what evidence.

1.3 Structure of the report

[TODO: One short paragraph describing Sections 2–5.]

2 Challenge & Scoping

2.1 Problem definition and design question

Given challenge (Challenge 1). *Cities are facing increasing heat stress due to climate change. Develop bioinspired strategies to reduce urban heat through building materials, façade design, or urban planning.*

The challenge as given spans three scales — material, building component and urban planning. Working on all three would make the project larger rather than more focused, which the systems view is explicitly meant to prevent. The design question is the instrument that makes this cut.

2.1.1 Candidate questions considered

Five candidates were formulated in the team and assessed against the criterion that a design question must be neither too broad nor too narrow: specific enough to define an intervention point, open enough to admit different solutions.

Table 1: Design questions considered, and why four were rejected.

#	Candidate	Assessment
1	How do we prevent heat build-up in urban areas?	<i>Too broad.</i> No place, no time, no intervention point — the equivalent of “How might we end hunger?”
2	How could we use the energy or heat build-up positively while lowering temperature?	Right idea, but unanchored: no location, and “positively” does not say for what. Refined into candidate 5.
3	How might we get the heat that Zurich’s existing buildings store during the day back out of the city before the next morning, without using energy?	Well formed and highly testable. Rejected only because it discards the team’s central idea of <i>using</i> the heat. Its night-time focus is retained in the selected question.
4	How can we prevent heat waves causing heat islands?	<i>Factually wrong.</i> Heat waves do not cause heat islands. The urban heat island is a permanent consequence of urban surface properties and geometry and exists on ordinary summer days; a heat wave makes it dangerous, it does not create it (Oke 1982).
5	How might we turn the heat that accumulates in a city into a useful resource, instead of reflecting or discarding it?	Good direction, but the reading “store summer heat for winter” is not verifiable within the project, and it has no biological support: organisms store chemical energy across seasons, not sensible heat. Narrowed into the selected question.

2.1.2 Selected design question

Design question (solution-neutral):

How might we let the heat that Zurich’s existing buildings absorb during a summer day drive its own removal, so that the hotter it gets, the more the building cools itself?

The question names no solution — no coating, no greening, no geometry, no water. It states the location (Zurich), the object (existing buildings, so retrofit rather than new build) and the operating season. What it adds to candidates 3 and 5 is the subordinate clause, and that clause is the actual content of the project: it demands a **self-regulating** response, in which the cooling power rises with the heat load instead of being fixed.

This matters because it distinguishes the project from the incumbent measure. A bright surface (Section 2.3) has a fixed optical response: it reflects the same fraction of radiation at 20 °C as at 40 °C, and it does nothing at all once the sun has set. A self-regulating surface behaves like a biological one — transpiration in a leaf is driven by the same solar energy that heats the leaf, so the cooling effort scales with the load, without a controller, a sensor or a pump.

It also resolves the team’s initial difficulty of having two systems of interest (“reduce heat” and “prevent heat”). These are not two systems. They are two functions of one system, coupled by the selected question: prevention acts during the day window, removal during the night window, and the self-regulating link connects them.

2.2 Context and constraints

Reference city: Zurich. All context values, benchmarks and engineering checks in this report refer to the City of Zurich (47.38° N, 8.54° E, approximately 400 m above sea level on the valley floor of the Limmat). Fixing one reference city is what makes the constraints below and the KPI targets in Table 8 checkable instead of generic. Zurich was chosen for four reasons:

1. The city has an adopted municipal heat-mitigation strategy, the *Fachplanung Hitze-minderung* (Stadt Zürich 2020a), with an implementation agenda (Stadt Zürich 2020c), a guidance leaflet for building owners (Stadt Zürich 2021) and three thematic maps (Stadt Zürich 2020b). The project therefore has a real client, a documented baseline and published target values instead of invented ones. Section 2.3 shows what this means for our design space.
2. The city publishes an open climate analysis and open geodata (Stadt Zürich n.d.), so the reference site can be selected from measured heat-load data instead of intuition.
3. MeteoSwiss operates long-term stations in and around the city (MeteoSwiss n.d.), which supplies the boundary conditions (G , T_a , wind speed, humidity) for the energy balance in Section 3.2.
4. Its building stock and morphology are representative of the Swiss Mittelland, so results transfer to comparable Swiss cities.

Reference site and users. We adopt the city’s own model area *Modellierungsgebiet 03 — Geschlossene Randbebauung* (closed perimeter block) as the reference site (Stadt Zürich 2020a, pp. 148–151). It consists of two outward-closed blocks of five storeys plus a roof structure enclosing private courtyards, in a dense district close to the centre; the street space is sealed from façade to façade and only isolated trees provide shade. The city’s own assessment is that the reduced air exchange and the lack of cold-air supply make this structural type overheat comparatively quickly.

Choosing this site has a methodical advantage: the Fachplanung already publishes modelled baseline and climate-optimised results for exactly this area, so our concept can be compared against a documented reference rather than against our own assumptions. **[TODO: Record address, façade orientations, roof area and year of construction of the specific building you pick inside this area.]**

The users are the residents of the upper floors, who carry the strongest heat load and usually have no mechanical cooling; the pedestrians in the street canyon, who receive radiation from the façade at eye level; the building owner, who decides on and pays for the retrofit; and the trades who install and later maintain the system.

When — the design case. The design case is not a single hot afternoon but a **multi-day heat wave**: five consecutive hot days ($T_{\max} \geq 30^\circ\text{C}$) without rain. This matters because the concept relies on a stored resource. A system that performs well on day one and is exhausted on day three has not solved the problem, and only a multi-day case puts the water reserve (F7) and

the reset behaviour of the heat store (F4) under real stress. The autonomy requirement in KPI 6 follows from this case. **[TODO: Confirm the duration and the frequency of such spells for Zurich from MeteoSwiss records, and state the station and reference period.]**

When — the critical windows. Within each day of that heat wave the design must perform in three windows, which follow one another without a gap because the surface never stops exchanging heat. Sunset in Zurich in high summer is around 21:00, sunrise around 06:00.

Table 2: Critical operating windows within one day of the design heat wave.

Window	What dominates	Why it is critical	KPIs
Load 11:00–18:00	Absorbed short-wave radiation. Clear-sky global irradiance of the order of 900 W/m^2 to 950 W/m^2 on a horizontal surface around solar noon (about 13:30 local time), air temperature up to roughly 33°C to 36°C .	The energy that has to be dealt with all night enters here. A sealed, sun-exposed surface can exceed 60°C (Stadt Zürich 2021). The city evaluates PET at 14:00, which lies inside this window, so our results stay comparable to Table 4.	2, 3, 4
Evening 18:00–24:00	Until sunset, low-angle sun on west-facing surfaces; after sunset, the start of the discharge of stored heat into the street canyon.	This is when residents are outdoors and then go to bed, so it is the window in which the effect is actually experienced. West façades receive their highest incidence angle here, which is why the window starts at 18:00 rather than at 20:00 — the two hours in between are among the hottest of the day at the surface, not a gap.	4, 5
Night 00:00–06:00	Long-wave emission to the sky and continued release of stored heat. Clearest sky conditions of the cycle.	Zurich’s urban heat island reaches 7 K at night against 1 K to 2 K by day (Stadt Zürich 2021), so this is the larger half of the problem and the half the incumbent measure does not address. It also sets a hard deadline: the heat store must be discharged and the surface reset by sunrise, or it saturates over the following days of the heat wave.	1, 5, 6

The remaining hours, 06:00–11:00, are not critical: the load is rising but still moderate. They are the margin the night window has to leave behind, and the state of the system at 06:00 is therefore reported as the reset condition for the next day.

Why the night window was extended. The team’s initial scoping put the second window at 20:00–24:00. Two things argued for changing it. First, the night-time reference instant of KPI 5 is 02:00, which the original window did not contain — a window must include the moments at which its own KPIs are evaluated. Second, tropical nights are defined over the whole night and the minimum temperature occurs shortly before sunrise, so a window ending at midnight would exclude the part of the night that defines the problem. Splitting the evening from the night also separates two genuinely different concerns: outdoor amenity while people are awake, and sleep quality plus the reset of the store afterwards. They need different KPIs, so they are treated as two windows.

Where on the envelope? A consequence of Zurich’s latitude. Fixing the reference city fixes the solar geometry, and that already narrows the design space. At 47.38° N the solar altitude at noon on 21 June is

$$h = 90^\circ - 47.38^\circ + 23.44^\circ \approx 66^\circ. \quad (1)$$

The direct beam therefore strikes a horizontal roof at an incidence factor of $\sin h \approx 0.91$, but a vertical south-facing façade at only $\cos h \approx 0.41$. In midsummer the flat roof receives more than twice the direct irradiance of a south façade at noon, and it also has a far larger sky view factor for night-time long-wave emission.

Both facts argue for the roof as a good place to *act*, and against it as the place where the problem is. The heat a roof stores is released *upward*, to a sky the roof can see; the heat a façade stores is released *sideways*, into the canyon, at the height of the people in it. Since Section 1 fixes the pedestrian-level outdoor space as the place where the effect has to appear, the system of interest is the **façade** — not because it receives more radiation, but because its heat is released where it does harm. Two further considerations support the same cut: a façade concept does not compete with photovoltaics for the roof area, which the constraints in Table 3 ask us to respect; and a systems view is meant to make a project more focused rather than larger.

The roof does not disappear from the project. It changes role: it is no longer a second surface to be treated, but the *interface with the good sky view* — the place where heat collected on the façade can be released, which is what the concept in Section 3.2 requires. **[TODO: Confirm the load profile with daily irradiation totals per orientation (Swiss Federal Office of Energy n.d.). East and west façades peak in the morning and late afternoon, so their load profile differs in shape, not only in size.]**

Constraints. Table 3 separates hard constraints (a violation disqualifies a concept) from soft ones (they weight the evaluation in Section 3.3). The constraints on installation and maintenance come from the team’s observation that the people who mount and service the system are part of the context; they are recorded here rather than as subsystems, because the system of interest contains components, not people.

2.3 State of the art: what the City of Zurich already does

Scoping is incomplete without knowing what the client already does. The Fachplanung Hitzeminderung defines thirteen courses of action (*Handlungsansätze*, HA 01–HA 13). Four act on the building envelope and therefore touch our project: HA 09 green roofs, HA 10 green façades, HA 11 high-albedo façade and roof materials, and HA 12 shading of the outdoor space close to the building (Stadt Zürich 2020a, pp. 120–131). HA 11 is the direct incumbent of our design question, and the city already recommends it to building owners (Stadt Zürich 2021).

Baseline values this fixes for us. The city’s model uses a solar reflectance of 0.2 for asphalt and 0.3 for un-greened façades, and classifies ≤ 0.30 as low and ≥ 0.80 as high albedo. Reported material values are slate 0.1, roof tiles 0.1–0.35, natural stone 0.2–0.35 and a white façade 0.5–0.9 (Stadt Zürich 2020a, pp. 114, 128–129). Our reference case and the target of KPI 2 in Table 8 are taken directly from these figures, so the comparison is made on the client’s terms rather than on ours.

The gap — and the design opportunity. Table 4 contains the central finding for this project. The incumbent measure works during the day but is close to ineffective at night: -0.1 K median air temperature, against an urban heat island that reaches 7 K at night. The reason is structural, not a question of tuning. A bright surface reduces what is *absorbed*; it provides no

Table 3: Context conditions and constraints.

Type	Class	Constraint
Energy	hard	No auxiliary electrical energy for the cooling function itself. This follows directly from the design question: the heat load must drive the response. Sensors used for validation are excluded.
Water	hard	No potable water. Only rainwater harvested on site or atmospheric moisture.
Existing building	hard	Retrofit onto an existing envelope. Load-bearing structure, floor plan and window openings are not redesigned.
Townscape	hard	The result must be acceptable in the Zurich townscape. Buildings may be listed; heritage protection, building law and neighbour rights can restrict visible changes (Stadt Zürich 2020a, p. 128).
Glare	hard	No specular glare towards streets, neighbouring flats or traffic. The city names glare as the main objection to bright envelopes (Stadt Zürich 2020a, p. 128), which favours diffuse over mirror-like surfaces.
Climate	hard	Must survive the Swiss annual cycle: frost to about -10°C , freeze-thaw cycles, driving rain, UV, and periods of several dry weeks in summer.
Winter	hard	Must not appreciably increase heating demand. A high solar reflectance also reduces useful winter solar gains; this trade-off must be quantified, not ignored.
Installation	soft	Must be installable by ordinary trades on an occupied building, ideally without full scaffolding. [TODO: state an installation-time target per m^2]
Maintenance	soft	Must remain functional with little maintenance. Soiling and the resulting cleaning cost are named by the city as a specific weakness of high-albedo envelopes (Stadt Zürich 2020a, p. 128). Service life target ≥ 20 years.
Solar energy	soft	Should not unnecessarily compromise roof- or façade-integrated photovoltaics; the city notes that bright envelopes reduce PV yield (Stadt Zürich 2020a, p. 128). State which envelope areas the concept claims.
Structural	soft	Limited additional dead load and build-up thickness on an existing envelope.
Embodied emissions	soft	The greenhouse gas embodied in the added build-up must be repaid by the avoided cooling demand within the service life. Measured in $\text{kg CO}_2\text{-eq l/m}^2$. [TODO: source a benchmark]
Economic	soft	The city expects higher cost for high-albedo envelopes (Stadt Zürich 2020a, p. 128). Target: cost per square metre in the range of a normal refurbishment. [TODO: find a benchmark figure and cite it]

Table 4: Modelled effectiveness of the incumbent measures, from the Fachplanung Hitzeminderung. Day values are the physiologically equivalent temperature (PET) at 14:00, 2 m above ground; night values are air temperature. “Range” is the distance over which the effect is measurable.

Measure	Median	Max.	Range	Source
HA 11 bright façade — day (PET)	−1.0 K	−1.8 K	2 m to 3 m	(Stadt Zürich 2020a, Tab. 11, 128)
HA 11 bright façade — night	−0.1 K	−0.2 K	1 m to 2 m	
HA 11 bright roof — day (PET)	−1.2 K	−2.1 K	2 m to 4 m	
HA 11 bright roof — night	−0.1 K	−0.3 K	1 m to 2 m	
HA 06 high-albedo ground surface instead of asphalt — day (PET)	−1.5 K	−2.8 K	2 m to 4 m	(Stadt Zürich 2020a, Tab. 5, 114)
HA 06 — night	−0.2 K	−0.5 K	2 m to 3 m	

path for releasing heat that is already stored in the construction, and its optical response is fixed — it does not do more when the load is higher, and it does nothing after sunset.

This is exactly what the selected design question targets. A bio-inspired concept earns its place here if it does at least one of the following:

1. deliver a night-time effect that HA 11 does not, by opening a removal path rather than only a prevention path;
2. scale its response with the load, so that the worst hours receive the strongest cooling;
3. resolve one of the constraints the city itself names for HA 11 — above all glare and soiling, which are precisely the points where biological surfaces are strong.

Two precedents outside the city’s own catalogue. The Fachplanung lists what the client does; it does not list what the field has already built. Two cases bear directly on our functions and are treated here as state of the art rather than as biological models. *StoColor Lotusan*, a micro-structured superhydrophobic façade paint modelled on the lotus leaf, has been on the market since 1999 and is stated by its manufacturer to resist algae and fungal growth *without biocide film protection* (Sto SE & Co. KGaA 1999). That is a commercial answer to precisely the soiling weakness the city names for HA 11, and it establishes that the physical route to F10 is not speculative. *HydroCanopy*, a façade system inspired by the mucous glands of frogs and by the density gradient of leaves in a tree canopy, combines solar regulation, ventilation, thermal inertia and evaporative cooling in one multi-layer build-up (AskNature 2026e). It is the closest published prior art to the concept direction of this project, which obliges us to state what our concept adds: a self-regulating link between load and cooling power (F9), and an effect in the night window.

[TODO: Note for Section 5: the running text on p. 128 of the Fachplanung quotes a maximum PET reduction of 2.3 K for a south-west facing façade, while Tab. 11 on the same page gives 1.8 K. Check the source, decide which value you use, and say so. Spotting this is worth more than quietly copying one of the two.]

2.4 System view and project boundary

Figure 1 places the challenge in its system hierarchy. The hierarchy prevents tunnel vision; the boundary drawn afterwards prevents scope explosion.

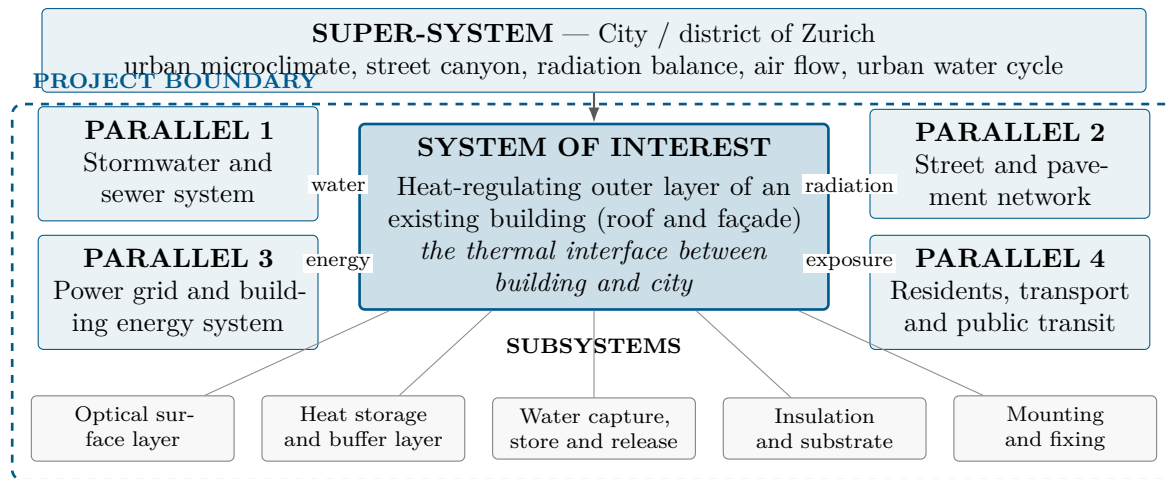


Figure 1: System view. The dashed line is the project boundary: everything inside it is designed and analysed by the team. Note that the subsystems are components and functions, not people — the trades who install and maintain the system appear as constraints in Table 3.

What each parallel system adds to the scoping. Listing parallel systems is not the point; what the comparison reveals is.

Table 5: What the comparison with each parallel system contributes.

Parallel system	What the comparison adds
Stormwater and sewer system	Rainwater is currently drained away as fast as possible. Seen from our system it is a cooling reserve that is being discarded — and during a heat wave it is the only water available under the “no potable water” constraint. This adds the functions <i>capture</i> , <i>retain</i> and <i>meter</i> water (F6–F8) and makes storage capacity a design variable rather than an afterthought.
Street and pavement network	The street is treated by the same client with the same logic (HA 06, Table 4), and it achieves a slightly larger daytime effect than the envelope. The comparison sets a benchmark we must beat, and it shows that the envelope’s advantage has to come from the night window, where the street performs no better.
Power grid and building energy system	The obvious way to “use” heat is to convert it. The comparison shows why that is a different project: conversion competes with photovoltaics for the same surface, and any seasonal store lies outside the envelope. It is therefore placed out of the boundary, while the milder reading of “use” — letting the heat drive its own removal — stays in.
Residents, transport and public transit	Defines where the effect has to be delivered: at pedestrian level in the street canyon and in the flats on the upper floors, not on the roof membrane. This is why PET at 2m above ground becomes a KPI, and it fixes the evening window 18:00–24:00, when people are outdoors and then sleeping.

What the systems view changed. *The systems view changed our project scope because it forced two corrections. First, we initially carried two systems of interest — reducing heat and preventing heat. The hierarchy showed that these are two functions of one system, and the selected design question couples them, so a single boundary can be drawn. Second, the comparison with the parallel systems revealed the night sky and the discarded rainwater as the two heat sinks available without energy input, and the pedestrian in the street canyon as the place where the effect must actually appear. This added the functions F2, F6–F8, the evening*

Table 6: Project boundary: what is deliberately in and out of the project.

IN the project	OUT of the project
Passive control of the short-wave and long-wave radiation balance of the envelope surface	Redesign of the building itself: structure, floor plan, windows, insulation thickness
The self-regulating link between heat load and cooling power — the core of the design question	Any actively controlled system: sensors, valves, pumps, chillers, mechanical ventilation
Heat storage and delayed release within the outermost build-up, including the night window	Seasonal heat storage, boreholes, heat pumps, thermal networks and electricity generation
Capture, retention and metered release of rainwater within the build-up	Municipal water supply, treatment and sewage; street trees and district-scale greening
Mounting on an existing substrate, installability and durability of the function	Permitting, heritage approval, business model, full life cycle assessment

operating window, and KPI 5 and 6 in Table 8.

2.5 Functions and functional diagram

Functions are formulated as solution-neutral verbs answering “what must the design *do*?”. They are stated at the scale of the designed object, not at the scale of the city: the design regulates heat at the surface it occupies, and the city-scale effect is the *goal*, not the function — in the same way that the function of polar bear fur is to insulate, not to make the Arctic habitable.

Why F9 is physically plausible. Every removal path already depends on surface temperature, but not equally strongly. Differentiating the three mechanisms with respect to surface temperature T_s gives their self-regulation gradients, in $\text{W}/(\text{m}^2 \text{K})$:

$$\text{radiative: } \frac{\partial P}{\partial T_s} = 4\varepsilon\sigma T_s^3 \approx 6 \quad (2)$$

$$\text{convective: } \frac{\partial P}{\partial T_s} = h_c \approx 10 \quad (3)$$

$$\text{latent: } \frac{\partial P}{\partial T_s} \propto \frac{de_s}{dT_s} \quad (\text{steep — } e_s \text{ roughly doubles per } 10 \text{ K}) \quad (4)$$

The evaporative path is by far the steepest, which is why transpiration, not radiation, is the mechanism that biology uses when a load must be tracked. **[TODO: Section 3.2: evaluate these three derivatives properly with your own assumptions and state the resulting ranking. This is the check that decides which mechanism carries F9.]**

2.6 Biologized questions

The functions are translated into questions to nature, which form the search strategy for Section 3.1. The challenge brief asks a further question directly — what can be learned from desert plants and animals about passive cooling — and the answer shapes where we looked: the organisms most challenged by our problem, and unfazed by it, live where solar load is extreme and water is scarce. That is why the models of Section 3.1 are disproportionately desert organisms, and why the two that transfer best are those whose strategy is a structure rather than a metabolic process.

- How does nature use the heat it receives to get rid of that same heat?
- How does nature make a response stronger when the load is stronger, without a controller?

Table 7: Function list. Main function, sub-functions and supporting functions.

ID	Function (verb)	Explanation / taxonomy link
<i>Main function</i>		
F0	Regulate heat gain and release at an urban surface	Modify / regulate physical state: manage temperature
<i>Sub-functions — prevention (day window)</i>		
F1	Reflect short-wave radiation	Prevent absorption of solar energy in the first place
F5	Shade a surface	Reduce the irradiated area, e.g. through surface geometry
<i>Sub-functions — removal (both windows)</i>		
F2	Emit long-wave radiation	Release heat to the sky, in particular in the atmospheric window
F3	Transfer heat to ambient air	Convective dissipation at the surface
F4	Store thermal energy temporarily	Buffer and shift the daily peak
<i>Sub-functions — water path</i>		
F6	Capture water	Collect rain or atmospheric moisture
F7	Retain water	Store it until it is needed during the heat peak
F8	Release water in a controlled way	Meter evaporation so the reserve lasts the whole heat wave
<i>Sub-function — the core of the design question</i>		
F9	Couple response strength to load	Increase removal (F2, F3, F8) as surface temperature rises, without external control
<i>Supporting functions</i>		
F10	Keep a surface clean	Maintain the optical function against soiling and algae
F11	Resist degradation	Withstand UV, frost, driving rain over the service life
F12	Attach to a substrate	Fix the build-up to an existing wall or roof, installable by ordinary trades

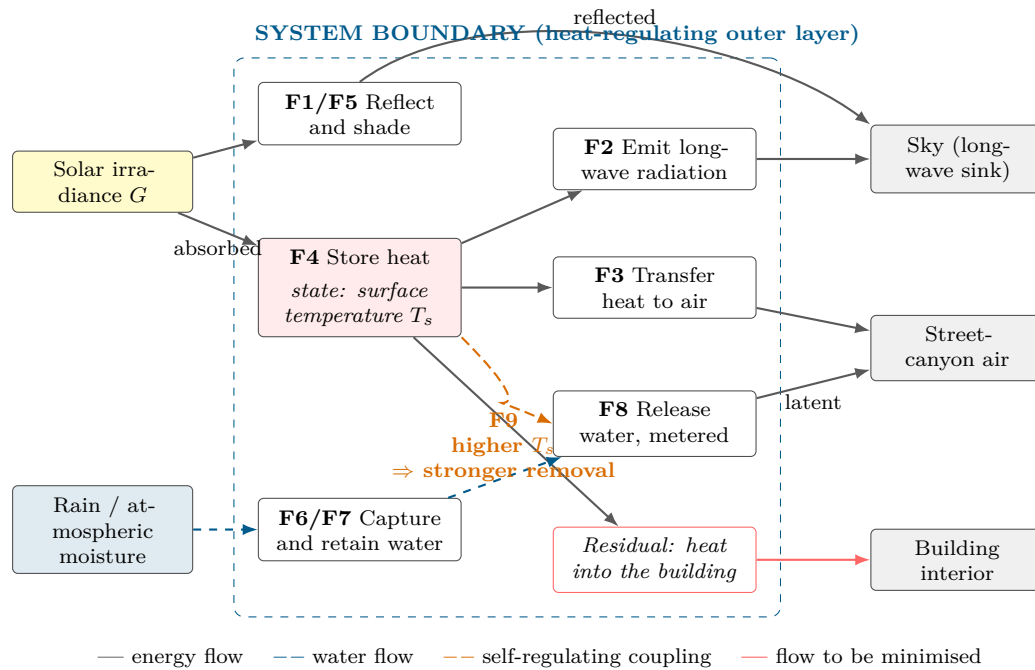


Figure 2: Solution-neutral functional diagram. Only functions and flows are shown, no components and no materials. The orange coupling is F9 and is what distinguishes this project from a fixed high-albedo surface: the state variable T_s feeds back into the strength of the removal functions. Supporting functions F10–F12 act on everything inside the boundary and are omitted for readability.

- How does nature release excess heat to the night sky?
- How does nature reflect intense solar radiation without pigments that bleach, and without a mirror finish?
- How does nature capture, store and meter out scarce water so that a reserve lasts a whole dry period?
- How does nature buffer a daily temperature extreme instead of resisting it?
- How does nature keep a functional surface clean without cleaning it?

2.7 Performance criteria and KPIs

The KPIs are derived from the functions, interfaces and constraints identified above, not invented independently of them. They are used twice: to guide the engineering checks in Section 3.2 and to compare the concepts in Section 3.3.

The reset condition. KPI 1 and KPI 5 are evaluated per day, but over the five-day design case a further condition applies: the surface temperature and the state of charge of the heat store at 06:00 must not drift upwards from one day to the next. A concept that cools well on the first day but saturates by the third fails the design case even if every single-day KPI is met. **[TODO: report the five-day trend in Section 3.2, not only day one]**

Why cost and embodied emissions are not KPIs. An earlier version of this table carried installed cost and embodied greenhouse gas as an eighth KPI. They have been moved into the constraints of Table 3, because a KPI must follow from a *function*, and neither of these does: they are conditions the design has to satisfy, not measures of how well it performs its job. The move also brings the list to seven, within the four to seven the task specification asks for.

Table 8: Performance criteria. Function $\rightarrow$ KPI $\rightarrow$ unit $\rightarrow$ target. Reference case: an existing Zurich façade or roof with a solar reflectance of about 0.3, the value the city uses for un-greened façades (Stadt Zürich 2020a, p. 114).

#	Function	KPI	Unit	Target / benchmark
1	F9 Couple response to load	Self-regulation gradient $\partial P_{\text{cool}}/\partial T_s$	W/(m ² K)	Must be clearly positive and larger than for a fixed high-albedo surface. The defining criterion of this project — it is the quantity the design question asks for. [TODO: set a target after the derivative check in Section 3.2]
2	F1 Reflect short-wave radiation	Solar reflectance α_{sol} (300–2500 nm)	–	≥ 0.80 , the threshold the city defines for “high albedo” (Stadt Zürich 2020a, p. 128). Baseline 0.3. Ultra-white laboratory paints reach ≈ 0.98 (Li et al. 2021).
3	F0 Regulate heat	Peak surface temperature reduction ΔT_s vs. reference, clear sky, 14:00	K	≥ 15 K. [TODO: confirm by calculation or measurement before claiming it]
4	F0 (day, people)	PET reduction at 2 m above ground, 14:00	K	Beat the incumbent. Since the system of interest is the façade, the binding figure is HA 11’s -1.0 K median for a bright façade; the -1.2 K for a bright roof is context, not our benchmark (Stadt Zürich 2020a, Tab. 11). This is the metric the client uses.
5	F2, F4 (evening / night)	Net cooling power at 02:00, clear sky; and heat discharged over the evening and night windows, 18:00–06:00	W/m ² ; W h/m ²	≥ 40 W/m ² instantaneous (Raman et al. 2014; Zhai et al. 2017), and a night-time air temperature effect better than the -0.1 K of HA 11 (Stadt Zürich 2020a, Tab. 11).
6	F6–F8 Water path	Potable water consumption; and days of autonomy on harvested rainwater	L/(m ² d); d	Potable water = 0 (hard constraint). Autonomy ≥ 5 days, so the reserve survives the five-day design heat wave of Table 2. [TODO: derive the required storage from the latent heat of vaporisation, 2.45 MJ/kg]
7	F10/F11 Durability	Retention of the cooling function after soiling and weathering	% of initial	≥ 90 % after three years. Soiling is the weakness the city names for high-albedo envelopes (Stadt Zürich 2020a, p. 128); aged cool roofs commonly lose a significant fraction of their initial reflectance (Santamouris 2014).

Two changes to the team’s original KPI list. Two of the KPIs originally proposed were reformulated, in both cases because of the rule that KPIs must follow from the functions and the system view.

Urban heat island intensity was dropped as a selection KPI. It is a district-scale quantity, and a single retrofitted building cannot move it measurably; keeping it would invite exactly the confusion between surface and air temperature described below. It remains in Section 1 as the context that motivates the project, and it is replaced operationally by KPI 4, which measures the same concern where it is actually attributable to our system.

Sustainability cost was not a measurable quantity as stated. It is replaced by the embodied-emissions constraint in Table 3, which has a unit, a benchmark and a decision rule.

One quantity deliberately not a KPI. The heat conducted through the envelope into the building is reported in Section 3.2 as a secondary result, because it follows from the same energy balance at no extra cost and matters to the building owner. It is not a selection KPI: indoor comfort and building services lie outside the project boundary (Table 6).

Important distinction. KPI 3 measures a *surface* temperature, KPI 4 an effect on people. A high-reflectance surface can be 20 K cooler than a dark one while the effect on the air temperature of the district is smaller by roughly two orders of magnitude (Santamouris 2014; Akbari et al. 2009). Conflating the two is the most common error in urban-heat-island projects, so the two are reported separately throughout this report.

Note on the targets. The targets above are aspirational benchmarks set during scoping. They are hypotheses, not results. Each is checked in Section 3.2 and either confirmed, revised or explicitly declared unverified. **[TODO: before hand-in: verify every cited benchmark value against its primary source]**

2.8 Relevant Life’s Principles

The six Life’s Principles are used here at the level the handbook defines them, with the sub-principle we actually claim named underneath. Listing a principle is not applying it; the assessment of how well the final concept achieves them is Section 4.

- **Be locally attuned and responsive** — sub-principles *harness freely available energy, leverage existing cycles*, and above all *the signal, the antennae that detect it and the response match*. This is the principle the design question is built on, and F9 is it expressed as a function. The two sinks we use — the night sky and the rainwater the city currently discards — are both freely available and both local.
- **Be resource efficient (material and energy)** — sub-principles *use low energy processes*, which is our hard energy constraint, and *a single design meets more than one need*, which is the tie-breaker we apply in Section 3.3: a build-up in which one layer carries several functions beats one that needs a layer per function.
- **Adapt to changing conditions** — sub-principles *adapt to temporal and spatial changes* (the design case is a five-day heat wave, not one hot afternoon), *withstand disturbance while maintaining function* (frost, freeze-thaw, driving rain, UV) and *maintain integrity through self-renewal* (soiling degrades the function; KPI 7).
- **Use life-friendly chemistry** — sub-principles *use a safe subset of the periodic table* and *the breakdown results only in harmless by-products*. This is not decorative for us: any biocide or stabiliser that leaches from the surface enters the rainwater we intend to collect and evaporate at pedestrian level, which is the place we are trying to improve.

- **Integrate development with growth** — claimed only in the limited sense of *built to shape*: a modular element sized to a façade grid produces no offcuts. The sub-principle *create more opportunities for life* is explicitly *not* claimed — a permanently damp textured surface invites exactly the algal growth that KPI 7 penalises, and naming that conflict is more honest than claiming the principle.
- **Evolve to survive** — **not claimed for the artefact**. A façade layer does not evolve across generations, and claiming it would be decoration. It is claimed for the *process*: this report builds on the incumbent rather than starting from zero, uses the world café for cross-pollination, and documents what was de-selected and why.

The umbrella principle, applied as a rule. *Create conditions conducive to life* opens with the requirement that a design be *optimised rather than maximised*. For us this is a concrete decision rather than a platitude: ultra-white coatings reach a solar reflectance of about 0.98 (Li et al. 2021), but KPI 2 targets ≥ 0.80 , the city’s own threshold, because further reflectance costs glare, winter solar gains and photovoltaic yield.

Quality check.

- ☐ Is our system of interest clearly defined? — Yes, and singular: the heat-regulating outer layer of an existing building.
- ☐ Is our project boundary manageable? — Yes, see Table 6. Seasonal storage and electricity generation are explicitly out.
- ☐ Are our functions solution-neutral? — Yes, verbs only; no components in Table 7 or Figure 2.
- ☐ Did the systems view reveal additional functions or constraints? — Yes: F2 and F6–F8 from the sky and the discarded rainwater, the evening window from the residents, and the installation and maintenance constraints.
- ☐ Are our most important functions measurable? — Yes, KPIs 1–8, each with a unit.
- ☐ Is the “How might we...?” question open to different solutions? — Yes: evaporative, radiative and geometric strategies all remain possible, and Section 3.2 must choose between them on evidence.

3 Bio-Inspired Design Process

The chain this section has to make visible is challenge → function → biology → abstraction → concept → engineering check → evaluation → iteration. Section 2 delivered the first two links. This section delivers the biology and the abstraction, and hands a set of transferable principles to the concept development in Section 3.2.

3.1 Discovering & Abstracting

3.1.1 Research path

Every search started from a verb in the function list of Table 7, never from an organism. This follows the rule that biological content is organised by the function it serves, which is also the structure of the Biomimicry Taxonomy and therefore of AskNature (Biomimicry Institute n.d.). Table 9 records which taxonomy branch was searched for which of our functions, and what it returned; the biologized questions themselves are in Section 2.6.

F9 is worth noting: it is the only one of our functions that does not sit in a single branch of the taxonomy. It is the *pairing* of “sense signals: temperature” with “process signals: respond to signals”, which is the same construction the Life’s Principles checklist uses when it asks that “the

Table 9: Research path. Each of our functions was translated into a taxonomy branch before any organism was considered.

Function	Taxonomy branch searched	Models it returned
F1, F5	Modify → physical state → light/colour; Protect from non-living threats → light	Desert snail, <i>Morpho</i> , <i>Cyphochilus</i> , tree bark, cactus ribs
F2, F3	Transform/convert energy → thermal, radiant; Distribute → energy	Elephant ears, elephant hairs, jackrabbit ears, Saharan silver ant
F4	Store → energy	Sea star, desert snail
F6–F8	Capture, absorb or filter → liquids; Store → liquids; Distribute → liquids	Leaf stomata, camel fur, honeybee water collection, bryophytes
F9	Sense signals → temperature <i>paired with</i> Process signals → respond to signals	Honeybee fanning thresholds, leaf stomata, pine cone, elephant hot spots
F10, F11	Protect from non-living threats → dirt/solids, light, ice	Lichen, lotus leaf

signal it sends, the antennae it uses to detect the signal, and its response match” (Baumeister et al. 2014). That the central function of our design question has to be assembled from two branches is itself a finding: it is why searching by organism would not have produced it.

Sources, in the order they were used. AskNature strategies and innovations to find candidates (Biomimicry Institute n.d.); the *Genius of Biome* report for the temperate broadleaf forest, which is the biome Zurich lies in, for models already translated to the built environment (Biomimicry 3.8 and HOK 2013); primary literature for the mechanisms; and commercial products where the transfer has already been made, which are treated as state of the art rather than as models. AskNature and AI-assisted search were used to *find* candidates only; every mechanism claim carried into this report is referenced to a paper or to the organisation that published the strategy.

Which lenses were applied. Of the five lenses in the method, two did most of the work. The *operating-condition lens* — intense solar radiation, surfaces above 60 °C, freeze–thaw, weeks without rain — led to the desert organisms. The *local lens* led to (Biomimicry 3.8 and HOK 2013), because Zurich sits in the temperate broadleaf forest biome and that report translates exactly this biome into building design. The *function lens* produced Table 9. The naturalist and ecological lenses contributed context but no model that survived selection.

3.1.2 Models found, and the selection decision

Two features of Table 10 are deliberate. First, the same organism appears three times — the elephant supplies ear-based transport, boundary-layer hairs and localised hot spots, and these are three different mechanisms serving three different functions, not one model counted three times. Second, five models were rejected *with a stated criterion* rather than quietly dropped. The criterion is the one at the head of the table, and it is worth stating explicitly because it explains an asymmetry in what survived: most cooling in nature is metabolic or behavioural, and an envelope has neither metabolism nor behaviour. The two models that transfer most directly — the snail shell and the cactus rib — are the two that are *structures rather than processes*.

Table 10: Biological models found, and the selection or de-selection decision. The criterion applied throughout: a model must be transferable to a *passive surface* that survives twenty years of the Swiss annual cycle.

Organism / model	Function	Mechanism	Decision
<i>Prevention — reduce what is absorbed</i>			
Desert snail <i>Sphincterochila boissieri</i>	F1, F4	Shell reflects $\approx 90\%$ of visible and $\approx 95\%$ of near-infrared light; in extreme heat the aperture is sealed with a calcified plate and the animal withdraws, leaving an insulating air space between body and hot ground (AskNature 2026k)	Selected. Passive, mineral, non-metabolic — the closest analogue to an envelope we found
<i>Morpho</i> butterfly	F1	Transparent chitin–air layers a few hundred nanometres apart diffract and recombine light; colour comes from geometry, and the disordered structure keeps the reflection diffuse rather than specular (AskNature 2026c)	Selected. Decouples appearance from absorptance, which is what the townscape constraint requires
White beetle <i>Cyphochilus</i>	F1	Brilliant whiteness from a disordered chitin network only $7 \pm 1.5\ \mu\text{m}$ thick, without pigment (Vukusic et al. 2007)	Selected as the achromatic case of the same principle
Saharan silver ant <i>Cataglyphis bombycina</i>	F1, F2	Triangular hairs scatter visible and near-infrared light and simultaneously raise emissivity in the mid-infrared (Shi et al. 2015)	Selected. The only model that does F1 and F2 in one structure
Cactus ribs	F5, F3, F2	Peaks shade the troughs; the cooler trough air absorbs heat and rises out, while the folds enlarge the emitting area (AskNature 2026d)	Selected. Geometry only — no material, no water, nothing to bleach
Tree bark	F1, F2	Reflects part of the incoming radiation while emitting efficiently; rough layering shades itself and traps insulating air (AskNature 2026a)	Selected. A dead outer layer protecting a living structure — the same role our build-up has
<i>Removal — get heat out again</i>			
Elephant ears	F2, F3	Warm blood is carried from the core into large thin ears; the surface releases it, and flapping raises the airflow (AskNature 2026h)	Selected. Separates the place of absorption from the place of release
Elephant skin hairs	F3	Sparse hairs reach through the still boundary layer so moving air reaches the skin; reported additional heat loss up to 23% (AskNature 2026l)	Selected. Directly relevant at the low wind speeds of a canyon
Jackrabbit ears	F9, F2	Vasodilation varies the blood flow to a thin radiating surface; at $\approx 30^\circ\text{C}$ air temperature the ears shed all excess heat, and the strategy exists to avoid spending water (AskNature 2026g)	Selected. The water-free route to a load-coupled response
Elephant skin hot spots	F9	Vasodilation in scattered patches rather than uniformly; the number of open “thermal windows” varies with the load (AskNature 2026f)	Selected. Regulation by active <i>area</i> rather than by material
<i>Water path and storage</i>			
Leaf stomata	F9, F8	Guard cells swell and shrink with their water content and so open and close the pore; evaporation itself pulls the water up, with no pump; leaf surfaces cool by 11 K	Selected. The clearest natural instance of F9

[TODO: Before hand-in: open every AskNature page cited in this table and replace it with the primary paper wherever one exists. Two are already identified: the silver-ant film reproduction is *Biologically inspired flexible photonic films for efficient passive radiative cooling*, PNAS 2020, and the adaptive infrared material is *A dynamic thermoregulatory material inspired by squid skin*, Nature Communications 2019. The course rule is explicit that AskNature finds candidates and the paper is the evidence.]

3.1.3 Mechanisms in depth

Three mechanisms are developed further here, one per functional family. They were chosen because each answers a question Section 2 left open.

Prevention — why reflection alone is not the strategy. The desert snail survives on sun-exposed ground at up to 50 °C without moving, without water, and without a metabolism that could help. Its shell reflects about 90 % of visible and about 95 % of near-infrared light, which is above the $\alpha_{\text{sol}} \geq 0.80$ that KPI 2 asks for and close to the ultra-white laboratory paints of Li et al. (2021) — achieved with calcium carbonate rather than a manufactured pigment. The part that matters for us is what happens next. In extreme conditions the animal seals the aperture and retracts into the second whorl, so an air space separates its tissue from the shell wall nearest the hot ground. Reflection reduces what is absorbed; the air space stops what is still absorbed from reaching what must be protected. For a retrofit this maps onto a construction that already exists: a reflective outer skin in front of a ventilated cavity. That matters for the night window, because the 7 K the city loses at night is heat released from mass that was charged during the day, and a decoupled outer skin charges far less of it.

Regulation — how a switch can behave like a controller. In a honeybee colony the temperature at which an individual worker begins to fan varies between individuals, and the variation is genetic: a colony has one queen but many fathers, so it contains patriline with different thresholds. As the nest warms, first a few bees fan, then more, then many, and the nest is held below about 36 °C with no central control at all. Each bee is a binary switch; the colony response is continuous. This resolves an objection that would otherwise disqualify every switching material considered in Section 3.2: a single threshold gives an almost infinite $\partial P / \partial T_s$ at the switching point and roughly zero either side of it, which is not the graded response KPI 1 asks for. A *distribution* of thresholds across many small units restores it. The same pattern appears in Biomimicry 3.8 and HOK (2013) as the “lots of littles” principle, arrived at independently.

Water — why metering matters more than harvesting. A leaf cools itself by 11 K to 17 K through pores whose two guard cells swell and shrink with their water content, opening and closing the pore as they do; the water is drawn up by the evaporation itself, with no pump anywhere (Biomimicry 3.8 and HOK 2013, pp. 62–65). The valve responds directly to heat and dryness, so the effort tracks the load. The camel supplies the complementary half. Its fur is insulation worn *in* the heat, and in the technical reproduction of that arrangement — an evaporating hydrogel underneath a heat-blocking but vapour-permeable aerogel — a sample was held 7 K below ambient for 200 h, five times longer than the wet layer alone (AskNature 2026i). Insulating the wet layer, rather than exposing it, is what converts a fixed volume of water into a duration. Since KPI 6 asks for five days of autonomy and Zurich supplies no rain during the design heat wave, metering is the binding problem and harvesting is not.

[TODO: Add one figure per mechanism above. Prompts for generating them are in Working documents/image_prompts.md; a generated figure is an illustration and must be captioned as schematic, and the AI use belongs in Appendix A.]

3.1.4 Abstraction into design principles

Table 11 keeps the three columns strictly separate, as the method requires: what was observed, what the observation abstracts to, and what it might become technically. Only the third column contains anything that could be built, and nothing in it has been selected yet — that is the work of Section 3.2.

Table 11: From biological observation to transferable design principle. The third column lists possibilities, not decisions.

Biological observation	Abstracted principle	Possible technical interpretation
The snail's shell reflects strongly <i>and</i> an air space separates it from the hot substrate	Reflect first, then decouple whatever still gets through with a layer of air	Reflective outer skin over a ventilated cavity; structural white instead of a pigment
<i>Morpho</i> colour comes from layer geometry, not pigment, and the disordered structure stays diffuse	Separate what a surface looks like from what it does with heat	Structurally coloured or near-infrared-selective finish that keeps the existing façade colour
Cactus peaks shade the troughs; cooler trough air absorbs heat and rises out	Use folded geometry for self-shading, more emitting area and better airflow	Ribbed or profiled façade and paving relief
Sparse hairs reach through the still air layer clinging to the skin	Break the boundary layer so moving air reaches the surface	Millimetre-scale texture on façade and pavement
The jackrabbit varies blood flow to a thin radiator; the elephant opens patches	Regulate by varying the <i>transport</i> to a radiating surface, and place the radiator only where conditions are best	Pumpless density-driven liquid loop from the canyon face to a parapet-level radiator with a good sky view
Guard cells open and close the pore in response to heat and dryness	Build the control into the opening, so the escape route widens with the load	Load-driven valve or wick, rather than a controlled one
Camel fur insulates <i>over</i> the evaporating skin	Cut the load first, then spend a little water on the remainder, and cover the wet layer so the reserve lasts	Absorbent layer beneath a vapour-permeable cover
Worker bees each switch at a different temperature	Spread many binary elements across a range of trigger points and the sum responds continuously	Any switching element built as many small units with a deliberate spread of switching temperatures
Pine cone scales bend because two bonded layers swell differently	Make moving parts from two layers that respond differently and let the environment actuate them	Bilayer shading element; hygroscopic or thermal actuation, no motor
The sea star charges on cold water sized by the previous hot exposure	Charge the store during the cool phase, sized by the load of the previous cycle	Night-charged store, with an explicit reset condition across the five-day case
Lichen roughness at several scales sheds water while staying breathable	Use geometry rather than chemistry to shed water and stay clean	Multi-scale textured finish; no biocide in the film

One principle, found three times. The jackrabbit and the elephant regulate by how much *area* is active, the bees by how many *units* have switched, and a bilayer shading element by how far it has *deployed*. All three modulate how many elements act rather than how strongly each one acts, and all three arrive at a graded response without a controller. That convergence from three unrelated organisms is the strongest abstraction this project has produced, and it is the one carried into every concept in Section 3.2.

3.1.5 Life's Principles evident in the selected strategies

The principles below are the ones the *selected models* actually exhibit; Section 4 assesses how well our final concept applies them, which is a different question.

- **Be locally attuned and responsive** — present in every model that couples its response to the local load, and named by the checklist in the form our F9 requires: the signal, the antenna that detects it and the response must match (Baumeister et al. 2014).
- **Be resource efficient (material and energy)**, sub-principle *use low energy processes* — none of the selected mechanisms consumes energy; the load itself drives them.
- **Adapt to changing conditions**, sub-principles *incorporate diversity* and *maintain integrity through self-renewal* — *Encelia* and the pine cone for the first, the lichen for the second.
- **Use life-friendly chemistry** — the lichen and the *Cyphochilus* structure both achieve with geometry what industry achieves with a compound that then has to be justified.

[TODO: Section 4 must *assess* these, not list them again. The rubric marks down a list.]

3.2 Creating: Engineering Concept Development

3.2.1 Ideation

[TODO: Run and document Crazy 8s: eight technically different ways of transferring the abstracted principles of Table 11. Vary the mechanism, not the shape, and do not evaluate while generating. Include a photo or scan of the sheets. Warning: four of the eight ideas in the lecture's own worked example (night flush, evaporative inlet, thermal buffer, shading skin) overlap our function list. If our eight reproduce that list we have copied the example instead of using our own abstractions.]

3.2.2 Morphological matrix

The rows are chosen, not inherited. Of our thirteen functions, a row belongs in the matrix only where alternative principles would lead to a meaningfully different concept. On that criterion F0 is excluded as the main function, F10 and F11 are requirements and engineering checks rather than sources of concept variants, and F12 is included only if the mounting really changes the architecture. That leaves five rows.

[TODO: Read at least three coherent combinations off this matrix and name them. A column is not a concept: check that the options combined are physically compatible. Then select two that are technically distinct — different physics, not different styling — and develop them below.]

3.2.3 Concept A — [TODO: name]

[TODO: Annotated drawing plus description: components, functional chain, flows, candidate materials, key technical parameters. Annotate inlet, channels, driving effect, outlet, heat path, materials, key dimensions and expected performance, and state explicitly which parts are inspired by biology and which are conventional engineering.]

3.2.4 Concept B — [TODO: name]

[TODO: Must be technically distinct from A. The four directions currently on the table are: a switching coating (prevention, no night effect); a self-deploying shade

Table 12: Morphological matrix. Rows are the five functions for which alternative principles produce genuinely different concepts; columns are the technical interpretations abstracted in Table 11.

Function	Option A	Option B	Option C
F1, F5 Reduce heat input	Structural white or near-infrared-selective finish	Ribbed, self-shading geometry	Deployable shading element
F2, F3 Move heat away	Broadband emitter with a good sky view	Boundary-layer texture for convection	Density-driven liquid loop to a parapet radiator
F4 Store and buffer	Thermal mass in the build-up	Latent store (phase change)	No added storage; decoupling only
F6–F8 Handle water	Absorbent layer under a vapour-permeable cover	Wick with load-driven release	No water path
F9 Couple response to load	Switching material, many units, spread thresholds	Load-driven valve or pore	Bilayer element that deploys with heat or moisture

(prevention plus an open night sky, acts at pedestrian level); an evaporative build-up with a water path (removal, metered per the camel bilayer); and a ribbed geometry (prevention and convection, no material and no water). The pairing with the widest physical separation is the deploying shade against the evaporative build-up.]

3.2.5 Engineering Canvas and critical questions

A critical question names a quantity, a threshold and a scale. Three follow directly from the abstractions of Section 3.1 and are carried forward as candidates:

1. Is the buoyancy in a pumpless density-driven loop sufficient to move the required heat flux over the height available in a five-storey block, and does the loop survive a Swiss winter?
2. At what wind speed must a self-deploying shading element furl, and what furls it? A thermal actuator cannot sense wind, so an independent and equally passive release is required before the concept is safe over a street.
3. Can a switching element be produced with a deliberate *spread* of switching temperatures, as the honeybee threshold model requires, or does the chemistry force a single sharp threshold?
4. **[TODO: Select two or three of these, or replace them once the concepts are fixed. They must be the assumptions that could change the design decision.]**

3.2.6 Engineering check 1: steady-state surface energy balance

[TODO: This check is prepared because it is the most direct way to test KPI 1 and KPI 3. Fill in your own numbers, state every assumption, and keep the units.]

For an opaque surface in steady state, the energy balance per unit area is

$$(1 - \alpha_{\text{sol}}) G + \varepsilon L_{\downarrow} - \varepsilon \sigma T_s^4 - h_c (T_s - T_a) - q_{\text{cond}} = 0, \quad (5)$$

where α_{sol} is the solar reflectance, G the global irradiance on the surface, ε the thermal emittance, $L_{\downarrow}$ the incoming long-wave radiation from the sky, $\sigma = 5.67 \times 10^{-8} \text{ W}/(\text{m}^2 \text{ K}^4)$ the Stefan–Boltzmann constant, T_s the surface temperature, T_a the air temperature, h_c the convective heat transfer coefficient and q_{cond} the heat conducted into the construction.

Table 13: Engineering Canvas. One per concept; the output of the canvas is the list of critical questions below it.

Field	Concept [TODO: A / B]
Function — what must it achieve?	[TODO:]
Biological mechanism	[TODO:]
Abstracted principle	[TODO:]
Technical principle — what is transferred?	[TODO:]
Architecture — components and interfaces	[TODO:]
Inputs / outputs — material, energy, information	[TODO:]
Requirements / KPIs	[TODO:]
Assumptions	[TODO:]
Uncertainties — what could invalidate it?	[TODO:]
Next check — what should we calculate or test?	[TODO:]

Table 14: Assumptions for the energy balance. Every value needs a source.

Quantity	Value used	Justification / source
G	[TODO:] W/m^2	[TODO: MeteoSwiss, clear-sky summer day; Table 2 gives $900 \text{ W}/\text{m}^2$ to $950 \text{ W}/\text{m}^2$ horizontal]
T_a	[TODO:] $^{\circ}\text{C}$	[TODO: Table 2: 33°C to 36°C in the load window]
h_c	[TODO:] $\text{W}/(\text{m}^2 \text{ K})$	[TODO: low wind speed in a street canyon]
$L_{\downarrow}$	[TODO:] W/m^2	[TODO: clear-sky estimate; state the correlation used]
α_{sol} reference	0.3	Value the city uses for un-greened façades (Stadt Zürich 2020a, p. 114)
α_{sol} concept	[TODO:]	[TODO: target ≥ 0.80 , KPI 2]
ε	[TODO:]	[TODO: state whether the concept changes emittance or only reflectance — these are separate questions]

Result and meaning. [TODO: Solve Eq. (5) for T_s for the reference and for the concept, report both, and state explicitly what the difference does and does *not* prove (see the note on KPI 3 in Section 2.7).]

3.2.7 Engineering check 2: [TODO: title]

[TODO: Options that are realistic within the week: (a) a simple outdoor measurement — identical plates with different surfaces, logged over a few hours; (b) a transient 1-D model of the build-up over a diurnal cycle; (c) a water balance — litres of rainwater per square metre per hot day against the latent heat of vaporisation of 2.45 MJ/kg, checked against the 200 h the camel bilayer achieved (AskNature 2026i); (d) a material and cost comparison. Report method, raw data, result and uncertainty.]

A scale check that applies to four of our models. Micro-scale effects do not automatically survive being made square metres large. The *Cyphochilus* whiteness, the *Morpho* colour, the lichen roughness and the stomatal valve all live at micrometre scale, and the failure mode is named explicitly in the method: upscaling can break the function. [TODO: State for each of the four whether the effect scales, and cite the evidence. For the lotus effect the answer is partly known, because a lotus-effect façade paint has been manufactured since 1999 (Sto SE & Co. KGaA 1999).]

3.3 Concept Evaluation & Iteration

Table 15: Decision matrix. Ratings 1–5, weights from Section 2.7. Every rating must be justified in the text by a calculation or a reasoned estimate.

Criterion	Weight	Concept A	Concept B	Evidence used
KPI 1 – Self-regulation gradient	[TODO:]	[TODO:]	[TODO:]	[TODO:]
KPI 2 – Solar reflectance	[TODO:]	[TODO:]	[TODO:]	[TODO:]
KPI 3 – Peak surface temperature	[TODO:]	[TODO:]	[TODO:]	[TODO:]
KPI 4 – PET at 2 m	[TODO:]	[TODO:]	[TODO:]	[TODO:]
KPI 5 – Night radiative cooling	[TODO:]	[TODO:]	[TODO:]	[TODO:]
KPI 6 – Water path and autonomy	[TODO:]	[TODO:]	[TODO:]	[TODO:]
KPI 7 – Durability of the function	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Feasibility	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Quality of the biomimetic transfer	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Evidence and uncertainty	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Weighted total		[TODO:]	[TODO:]	

The last three rows are not KPIs and are deliberately separate from them. They carry the questions a KPI table cannot answer: whether the concept can be built, whether it genuinely transfers the biology or merely resembles it, and how well supported the ratings above are. The third of these is the one that prevents a confident-looking table from concealing guesswork.

3.3.1 Review and feedback

[TODO: World café peer feedback, instructor feedback, and AI used as a critical reviewer. For every AI comment record the decision — accept, reject or investigate]

— and the justification. A documented rejection is worth more than a list of accepted comments, because it shows judgement. This belongs in Appendix A as well.]

3.3.2 Selection and Final Concept V2

[TODO: State which concept, or which justified hybrid, was selected, and describe V2 in detail. The minimum evidence of iteration is one named weakness or uncertainty in the earlier concept and a documented response to it in V2.]

Table 16: Final output of the creating and evaluating phase: Concept V1a/b → engineering review → evidence → Concept V2.

What changed	Why it changed	Evidence that informed the change
[TODO:]	[TODO:]	[TODO:]
[TODO:]	[TODO:]	[TODO:]

3.3.3 Remaining uncertainties

[TODO: What is still unproven, and what should be tested or developed next. The scale question above and the wind release for a deploying element are the two that are already known to be open.]

4 Sustainability & Life's Principles

4.1 Application of Life's Principles to the final concept

[TODO: Take the principles from Section 2.8 and assess each one honestly: how well does the final concept apply it, and where does it fall short? A table with a short assessment per principle works well. Listing principles without assessing them is explicitly marked down in the rubric.]

4.2 Environmental benefits, limitations and trade-offs

[TODO: Quantify where possible. The trade-offs that are already visible for this challenge:]

- **Summer vs. winter:** a high solar reflectance reduces summer heat gain but also reduces useful passive solar gains in winter. [TODO: estimate both and give the net balance for the Swiss climate]
- **Cooling vs. water:** evaporative cooling costs roughly 2.45 MJ per kilogram of water evaporated. [TODO: convert this into litres per square metre per hot day and check it against the rainwater actually available on site — during a heat wave it will not rain, which is precisely the point of the storage function F7]
- **Local vs. global:** the concept moves heat out of the street canyon; state clearly whether it reduces total energy demand or only redistributes heat.
- **Materials:** [TODO: pigments, binders, biocides against algae, end of life, separability of the layers, recyclability]

4.3 Influence of sustainability considerations on the design

[TODO: Name at least one concrete design decision that was changed because of a sustainability consideration — otherwise this section reads as an add-on.]

5 Discussion & Conclusion

5.1 Contribution of the biomimicry process

[TODO: Be concrete and honest. Which idea came specifically from the biological model, and which would a conventional engineering approach have produced anyway? A well-argued “this part was conventional” is worth more than an overstated claim.]

5.2 Strengths and weaknesses of the final concept

[TODO:]

5.3 Assessment against the challenge and the KPIs

[TODO: Go back through KPIs 1–7 from Table 8 and state for each one whether it was met, missed, or could not be assessed with the available evidence.]

5.4 Innovative character and limits of the evidence

[TODO: State clearly what has been demonstrated (calculation, measurement, literature) and what remains an assumption.]

6 References

- Akbari, Hashem, Surabi Menon, and Arthur Rosenfeld (2009). “Global cooling: Increasing world-wide urban albedos to offset CO₂”. In: *Climatic Change* 94.3, pp. 275–286. DOI: [10.1007/s10584-008-9515-9](https://doi.org/10.1007/s10584-008-9515-9).
- AskNature (2026a). *Bark Keeps Surface Cool Under the Sun*. Biomimicry Institute. URL: <https://asknature.org/strategy/bark-keeps-surface-cool-under-the-sun/> (visited on 09/02/2026).
- (2026b). *Body Buffers Thermal Variations*. Biomimicry Institute. URL: <https://asknature.org/strategy/body-buffers-thermal-variations/> (visited on 09/02/2026).
- (2026c). *Butterflies Hack Light Waves to Produce Brilliant Color*. Biomimicry Institute. URL: <https://asknature.org/strategy/wing-scales-cause-light-to-diffract-and-interfere/> (visited on 09/02/2026).
- (2026d). *Cactus Ribs Keep Them Cool*. Biomimicry Institute. URL: <https://asknature.org/strategy/shape-shades-and-enhances-heat-radiation/> (visited on 09/02/2026).
- (2026e). *Cooling Façade Inspired by Frogs and the Leaf Canopy*. Biomimicry Institute, Innovation. URL: <https://asknature.org/innovation/evaporative-cooling-facade-inspired-by-frogs-and-the-leaf-canopy/> (visited on 09/02/2026).
- (2026f). *Hot Spots on Skin Keep Elephants Cool*. Biomimicry Institute. URL: <https://asknature.org/strategy/skin-fine-tunes-internal-temperature/> (visited on 09/02/2026).
- (2026g). *How Blood Flow Keeps Jackrabbits Cool*. Biomimicry Institute. URL: <https://asknature.org/strategy/how-blood-flow-keeps-jackrabbits-cool/> (visited on 09/02/2026).
- (2026h). *Large Ears Aid Cooling*. Biomimicry Institute. URL: <https://asknature.org/strategy/large-ears-aid-cooling/> (visited on 09/02/2026).

- AskNature (2026i). *Passive Cooling System Inspired by Camels*. Biomimicry Institute, Innovation. Bilayer hydrogel–aerogel cooler developed at MIT. **[TODO: trace to the primary publication]**. URL: <https://asknature.org/innovation/passive-cooling-system-inspired-by-camels/> (visited on 09/02/2026).
- (2026j). *Pine Cones Open and Close in Response to Weather*. Biomimicry Institute. URL: <https://asknature.org/strategy/pine-cones-open-and-close-in-response-to-weather/> (visited on 09/02/2026).
- (2026k). *Shell Protects From Heat*. Biomimicry Institute. URL: <https://asknature.org/strategy/shell-protects-from-heat/> (visited on 09/02/2026).
- (2026l). *Small Leaves Buffer Insect Eggs From Heat*. Biomimicry Institute. Used for the boundary-layer argument. **[TODO: the 23 % figure comes from the University of Notre Dame biomechanics blog — trace it to the underlying study]**. URL: <https://asknature.org/strategy/small-leaves-buffer-ambient-temperature> (visited on 09/02/2026).
- (2026m). *Varying Response Thresholds Aid Hive Thermoregulation*. Biomimicry Institute. URL: <https://asknature.org/strategy/varying-response-thresholds-aid-hive-thermoregulation/> (visited on 09/02/2026).
- Baumeister, Dayna et al. (2014). *Biomimicry Resource Handbook: A Seed Bank of Best Practices*. Missoula, MT: Biomimicry 3.8.
- Biomimicry 3.8 and HOK (2013). *Genius of Biome: Temperate Broadleaf Forest*. Design strategies of the temperate broadleaf forest biome translated to the built environment. Local copy in the course folder “Day 2 Discovering and abstracting”. Biomimicry Group, Inc. and HOK Group, Inc.
- Biomimicry Institute (n.d.). *AskNature*. <https://asknature.org/>. Accessed **[TODO: date]**.
- Ehleringer, James and Harold A. Mooney (1978). “Leaf hairs: Effects on physiological activity and adaptive value to a desert shrub”. In: *Oecologia* 37.2, pp. 183–200. DOI: [10.1007/BF00344990](https://doi.org/10.1007/BF00344990).
- Helms, Michael, Swaroop S. Vattam, and Ashok K. Goel (2009). “Biologically inspired design: Process and products”. In: *Design Studies* 30.5, pp. 606–622. DOI: [10.1016/j.destud.2009.04.003](https://doi.org/10.1016/j.destud.2009.04.003).
- Li, Xiangyu et al. (2021). “Ultrawhite BaSO₄ paints and films for remarkable daytime subambient radiative cooling”. In: *ACS Applied Materials & Interfaces* 13.18, pp. 21733–21739. DOI: [10.1021/acsami.1c02368](https://doi.org/10.1021/acsami.1c02368).
- Martins, António F. et al. (2018). “Locally-curved geometry generates bending cracks in the African elephant skin”. In: *Nature Communications* 9, p. 3865. DOI: [10.1038/s41467-018-06257-3](https://doi.org/10.1038/s41467-018-06257-3).
- MeteoSwiss (n.d.). *Climate data of the automatic monitoring network*. Federal Office of Meteorology and Climatology MeteoSwiss. Zurich stations; access via IDAWEB / opendata. **[TODO: name the exact stations and reference period you use]**.
- Oke, T. R. (1982). “The energetic basis of the urban heat island”. In: *Quarterly Journal of the Royal Meteorological Society* 108.455, pp. 1–24. DOI: [10.1002/qj.49710845502](https://doi.org/10.1002/qj.49710845502).
- Raman, Aaswath P. et al. (2014). “Passive radiative cooling below ambient air temperature under direct sunlight”. In: *Nature* 515.7528, pp. 540–544. DOI: [10.1038/nature13883](https://doi.org/10.1038/nature13883).
- Santamouris, M. (2014). “Cooling the cities — A review of reflective and green roof mitigation technologies to fight heat island and improve comfort in urban environments”. In: *Solar Energy* 103, pp. 682–703. DOI: [10.1016/j.solener.2012.07.003](https://doi.org/10.1016/j.solener.2012.07.003).
- Shi, Norman Nan et al. (2015). “Keeping cool: Enhanced optical reflection and radiative heat dissipation in Saharan silver ants”. In: *Science* 349.6245, pp. 298–301. DOI: [10.1126/science.aab3564](https://doi.org/10.1126/science.aab3564).
- Stadt Zürich (Jan. 20, 2020a). *Fachplanung Hitzeminderung*. Programm Klimaanpassung. Modelling by GEO-NET Umweltconsulting GmbH. File: [fp-hitzeminderung-bericht.pdf](#).

- Zürich: Grün Stadt Zürich. URL: https://www.stadt-zuerich.ch/fachplanung_hitzeminderung.
- Stadt Zürich (2020b). *Teilpläne zur Fachplanung Hitzeminderung: Hitzeminderung, Kaltluftsystem, Entlastungssystem*. Stadt Zürich. Files: [teilplan-hitzeminderung.pdf](#), [teilplan-kaltluftsystem.pdf](#), [teilplan-entlastungssystem.pdf](#).
- (Jan. 20, 2020c). *Umsetzungsagenda 2020–2023 zur Fachplanung Hitzeminderung*. Programm Klimaanpassung. File: [fp-hitzeminderung-umsetzungsagenda.pdf](#). Zürich: Umwelt- und Gesundheitsschutz Zürich.
- (2021). *Merkblatt Hitzeminderung in der Stadt Zürich*. Umwelt- und Gesundheitsschutz Zürich, Fachbereich Umweltpolitik. Planning aid for building and land owners. File: [merkblatt-fp-hitzeminderung](#).
- (n.d.). *Open Data Katalog der Stadt Zürich*. <https://data.stadt-zuerich.ch/>. Building footprints, tree cadastre, climate layers. Accessed **[TODO: date]**.
- Sto SE & Co. KGaA (1999). *StoColor Lotusan — exterior coating with Lotus-Effect technology*. Product documentation. Micro-structured superhydrophobic façade paint; the manufacturer states resistance to algae and fungi without biocide film protection. **[TODO: verify the durability claims against independent literature before citing them as fact]**. URL: <https://www.sto.com/biomimetics/en/biomimetics/lotusan/index.html> (visited on 09/02/2026).
- Swiss Federal Office of Energy (n.d.). *sonnendach.ch — solar irradiance per roof surface*. <https://www.sonnendach.ch/>. Irradiation, orientation, tilt and usable area per roof. Accessed **[TODO: date]**.
- Turner, J. Scott and Rupert C. Soar (2008). “Beyond biomimicry: What termites can tell us about realizing the living building”. In: *Proceedings of the First International Conference on Industrialized, Intelligent Construction (I3CON)*. Critical re-assessment of the termite-mound ventilation narrative — read before citing the Eastgate Centre example. Loughborough University.
- Vukusic, Pete, Benny Hallam, and Joe Noyes (2007). “Brilliant whiteness in ultrathin beetle scales”. In: *Science* 315.5810, p. 348. DOI: [10.1126/science.1134666](https://doi.org/10.1126/science.1134666).
- Zhai, Yao et al. (2017). “Scalable-manufactured randomized glass-polymer hybrid metamaterial for daytime radiative cooling”. In: *Science* 355.6329, pp. 1062–1066. DOI: [10.1126/science.aai7899](https://doi.org/10.1126/science.aai7899).

Appendix A – AI Use Statement

Relevant AI-supported activities are documented below. Not every prompt is listed; the focus is on how AI contributed to the work and how important outputs were verified. AI output is not used as evidence for factual claims.

Appendix B – Optional Supporting Material

[TODO: Additional calculations, extended concept drawings, raw measurement data or prompt excerpts where useful.]

Table A.1: AI use statement.

AI tool	Used for	How the output influenced the project	How important output was verified
AskNature Chat	Discovering biological strategies for functions F1–F8	[TODO: e.g. generated candidate organisms for further research]	[TODO: AskNature profile + peer-reviewed paper]
[TODO: LLM assistant]	[TODO: e.g. structuring the scoping, drafting the LaTeX report template, critical review of the concepts]	[TODO: e.g. produced the first draft of the function list, which the team revised]	[TODO: e.g. every biological and physical claim checked against the cited primary sources; benchmark values recomputed by hand]